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**Master Thesis**

**Bayesian Additive Regression Trees  
for Causal Inference**

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## Abstract

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## Abstract

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# 1 Introduction

A central goal across many research fields is to understand the effect that an intervention or treatment has on an outcome. Examples include medicine (e.g. whether a medication leads to an improvement in patient health), policy making (e.g. whether stricter traffic rules lead to fewer accidents) or social sciences (e.g. whether tutoring leads to better test scores). In such settings, the objective is not merely to identify an association between the two variables treatment and outcome. Instead, the goal is to establish and quantify a causal relationship, where the treatment has an effect on the outcome. To do so, researchers typically compare a treatment group, which receives the intervention, with a control group, which does not. As estimating causal effects differs from identifying associations, predictive models alone are generally not sufficient in this context (Holland, 1986). Instead, specific methods which take the distinction between correlation and causation into account are needed.

Over the past decades, a wide range of methods for causal effect estimation have been developed. Recently, machine learning methods have become popular in causal inference as they can model complex relationships between covariates, treatment assignment and outcome and do not require explicit model specification (Feuerriegel et al., 2024). One of those methods is Bayesian Additive Regression Trees (BART), a flexible sum-of-trees model which was originally developed as a predictive model by Chipman et al. (2010) and has been adapted for the purpose of causal inference by J. L. Hill (2011). In a variety of studies, causal BART has been shown to yield reliable treatment effect estimates and often performs favorably compared to other methods (Caron et al., 2022; Hahn et al., 2020; Miratrix et al., 2025).

Despite its growing popularity, the literature on causal BART has not yet sufficiently studied two important aspects. First, causal BART can be implemented using different meta-learners. Two common approaches are S-learners and T-learners (J. L. Hill, 2011; Künzel et al., 2019). The S-learner estimates treatment effects with a single model for both treated and untreated units. The T-learner uses two separate models for treated units and untreated units. Furthermore, propensity scores can be included in the S-learner as an additional covariate (Hahn et al., 2020). In existing studies, often only one model is evaluated while different learners are rarely compared systematically. Second, BART

can be finetuned through several hyperparameters which control model complexity and regularization. Existing default values have been established by Chipman et al. (2010). Researchers often adopt those defaults instead of examining whether they are appropriate in different settings. Notably, the original paper which introduced causal BART (J. L. Hill, 2011), did not assess whether the established hyperparameters are also optimal for causal inference. Hyperparameter tuning can often substantially affect performance of causal models (Machlanski et al., 2023). Therefore, neglecting hyperparameter tuning when comparing causal effect estimation models may result in misleading conclusions.

To address these gaps, this thesis investigates the following research questions:

1. How does the performance of different causal BART learners compare?
2. How do BART-based approaches compare to other causal effect estimation methods?
3. To what extent can hyperparameter tuning improve model performance, and can tuning alter which method performs best?

To answer these questions, six causal effect estimation methods are evaluated. Four of these are BART models: an S-learner, a T-learner and two S-learners that include propensity scores. One of them estimates the propensity scores using a logistic regression, the other using a BART model. These models are compared with two other methods: Augmented inverse probability weighting (AIPW) and causal forests. AIPW is a simple parametric yet doubly robust approach, which combines propensity score weighting with an outcome regression model, while causal forests are a tree-based method like BART and specifically designed to detect heterogeneous treatment effects.

The methods are evaluated on 16 synthetic and semi-synthetic data-generating processes from the Atlantic Causal Inference Conference (ACIC) Data Challenge of 2019. The analysis focuses on binary treatment assignment, continuous outcomes and average treatment effect estimation. The performance of the models is assessed using the root mean squared error (RMSE), coverage and confidence interval length, both overall and across different dataset characteristics. Additionally, hyperparameter tuning is performed on four selected datasets to evaluate the extent to which tuning can significantly improve estimation performance and influence how models perform relative to each other.

### 1.1 Thesis outline

The remainder of this thesis is structured as follows. Chapter 2 introduces BART as a predictive machine learning model. Chapter 3 presents the foundations of causal inference,

including the potential outcomes framework and key challenges in estimating causal effects. Chapter 4 reviews general concepts of causal machine learning methods, discusses how BART can be adapted for treatment effect estimation, and introduces the causal forest and AIPW model. Chapter 5 describes the research design, data, and methodology used in the empirical analysis. Chapter 6 presents the empirical results, which are discussed in Chapter 7. Finally, Chapter 8 concludes the thesis and outlines directions for future research.





## 2 Bayesian Additive Regression Trees

### 2.1 Classification and Regression Trees

The methodological foundation of Bayesian Additive Regression Trees (BART) are Classification and Regression Trees (CART), introduced by Breiman et al. (1984), which ~~has~~ <sup>we have</sup> become a widely used approach for predictive machine learning tasks. The objective of CART models is to make predictions on an outcome variable  $Y$  based on a set of covariates  $X = (X_1, X_2, \dots, X_p)$ . When the outcome variable is categorical, the model is referred to as a classification tree; when it is continuous, it is referred to as a regression tree. In this thesis BART is used for regression tasks, therefore the following discussion focuses primarily on this setting. The description of the algorithm in this chapter is based on ~~Hastie et al.~~ <sup>parent</sup> (2009, Chapter 4).

A regression tree consists of nodes that are connected by branches, resulting in a hierarchical tree-like structure (Fig. X TODO). Its core idea is to recursively partition the covariate space into disjoint regions such that observations within each region are as similar as possible with respect to the outcome variable. The top node is referred to as the root node, while nodes without further splits are called terminal nodes or leaves. The splits are done at the internal nodes based on a splitting rule.

A splitting rule is defined by one of the covariates together with a split value. For numerical covariates, a threshold is chosen. Observations with values smaller than or equal to the threshold are assigned to one branch, while observations with values greater than the threshold are assigned to the other branch. For categorical covariates, categories are partitioned into either of the two branches. To choose a splitting rule, the algorithm compares all different covariates and split values and selects the rule that optimizes a predefined criterion. In regression trees, a common criterion is the residual sum of squares.

Tree construction begins at the root node, using the full training data set. After the first split is selected, the observations are partitioned into two subsets according to the splitting rule. The process is repeated at each of the resulting nodes, considering only the subset of data that was sorted into the node. In principle, this could continue until each observation occupies its own terminal node, resulting in a perfect fit to the training data. Such deep trees generally do not generalize well to new data because they fit the random variation in

the training data too closely, which results in low bias but high variance.

To prevent overfitting, it is generally advised to restrict tree growth to a certain depth. Common strategies include defining a minimum number of observations in each node or pruning. In pruning, initially a deep tree is grown and then branches are removed, if they do not improve the predictive performance of the tree sufficiently.

When the tree structure has been finalized, prediction values are assigned to the terminal nodes. Typically, the mean outcome value of the observations in each node is calculated. To make predictions for a new observation, it is passed through the tree according to the splitting rules until it reaches a leaf node. The prediction assigned to that leaf then serves as the predicted outcome.

Regression trees are able to capture nonlinear relationships and complex interactions between covariates without requiring explicit model specification. However, single trees are often unstable, as small changes in the training data can lead to substantially different tree structures. Therefore, ensembles of multiple trees are often used, such as BART (Chipman et al., 2010).

## 2.2 Bayesian Additive Regression Trees

Bayesian Additive Regression Trees, first introduced by (Chipman et al., 2010), build upon the concept of CART. The goal remains the same: predicting an outcome  $Y$  from a set of covariates  $X$ . Unlike standard regression trees, BART is a sum-of-trees model, meaning not one but multiple trees are grown and the final prediction is obtained by summing the predictions of the individual trees.

The following description in this section is based on Chipman et al. (2010). The model is defined as

$$Y = \sum_{j=1}^m g(x; T_j, M_j) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2) \quad (2.1)$$

where  $m$  is the number of trees and each function  $g(x; T_j, M_j)$  represents a single regression tree. Each tree is defined by its structure  $T_j$ , which encapsulates the set of internal nodes with their splitting rules, and a set of terminal node parameters  $M_j = \{\mu_{j1}, \mu_{j2}, \dots, \mu_{jb_j}\}$ . Each  $\mu_{jk}$  is the prediction assigned to the respective terminal node.

For a given observation with covariates  $x$ , the output of each individual tree is the parameter associated with the terminal node reached by  $x$  based on the splitting rules. The conditional expectation  $E[Y | x]$  is the sum of the contributions of all trees. The model



assumes Gaussian residual variation represented by the error term  $\varepsilon$ . Thus, the full model can be characterized by the parameters  $(T_1, M_1), \dots, (T_m, M_m), \sigma$ .

BART employs a Bayesian framework by placing priors on the tree structures, the terminal node parameters and the error variance, as well as by using a Gibbs sampler for posterior inference. This results in a posterior distribution over the model parameters, which in addition to providing point estimates, allows for uncertainty quantification.

Due to its structure, BART is a flexible and robust model. It can naturally handle continuous and categorical covariates. As a nonparametric model, BART does not require assumptions about the functional form of the relationship between the covariates and the outcome. The ensemble of trees is able to capture not only linear relationships, but also interactions and more complex, nonlinear patterns. Regularization is induced by choosing priors, which constrain each individual tree to only explain a small portion of the variation in the outcome. This helps prevent overfitting and leads to more stable predictions. Compared to single tree models, the variance of BART models is typically lower and it is better able to approximate smooth functions. BART has been shown to scale well to high-dimensional settings and is relatively robust to irrelevant covariates, which tend to be selected less frequently during tree construction.

## 2.3 Prior Specification

Employing a Bayesian framework, BART requires the specification of prior distributions over all model parameters. These parameters include the tree structure  $T_j$ , the terminal node parameters  $M_j$  and the error variance  $\sigma^2$ . The priors are chosen in such a way that shallow trees are favored, with the goal of regularizing the influence of individual trees. Various prior specifications have been proposed in the literature, and through empirical evaluation default priors have been established. The joint prior distribution over the model parameters is given by

$$p((T_1, M_1), \dots, (T_m, M_m), \sigma^2) \quad (2.2)$$

Specifying a prior on this joint distribution directly would be impractical. However, the prior can be decomposed into its components. Assuming independence between the trees and independence of  $\sigma^2$ , the joint prior can be written as

$$p((T_1, M_1), \dots, (T_m, M_m), \sigma^2) = \left[ \prod_{j=1}^m p(T_j, M_j) \right] p(\sigma^2) \quad (2.3)$$

Applying the product rule for joint distributions yields:

$$\left[ \prod_{j=1}^m p(T_j, M_j) \right] p(\sigma^2) = \left[ \prod_{j=1}^m p(M_j | T_j) p(T_j) \right] p(\sigma^2) \quad (2.4)$$

Furthermore, conditional independence between the terminal node parameters within each tree is assumed, resulting in

$$p(M_j | T_j) = \prod_{k=1}^{b_j} p(\mu_{jk} | T_j) \quad (2.5)$$

where  $b_j$  denotes the number of terminal nodes in tree  $j$ . Now, instead of placing a prior on the full joint distribution, priors can be placed on  $p(T_j)$ ,  $p(\mu_{jk} | T_j)$ , and  $p(\sigma^2)$  separately.

### 2.3.1 Tree structures

The prior on the tree structure consists of three components. The first component specifies the probability that a node splits at depth  $d$ :

$$\Pr(\text{node splits at depth } d) = \alpha(1 + d)^{-\beta}, \quad \alpha \in (0, 1), \beta \in [0, \infty) \quad (2.6)$$

where  $\alpha$  is the base split probability and  $\beta$  is the rate at which the splitting probability decreases with depth. A higher  $\alpha$  encourages more splits, meaning the tree will grow larger. Through a higher  $\beta$ , this probability decreases more quickly with increasing depth. (Chipman et al., 2010) recommend  $\alpha = 0.95$  and  $\beta = 2$ . These values encourage splits near the root and cause the splitting probability to decrease rapidly, which favors shallow trees overall.

The second component is the probability that a covariate  $x_j$  is selected for a splitting rule at a node. The default prior is a discrete uniform distribution over all covariates. The third component specifies the prior distribution over the splitting values of each covariate  $x_j$ . This prior is usually a uniform distribution over the observed available split points of  $x_j$ .

### 2.3.2 Terminal node parameters

Typically,  $Y$  is rescaled to approximately lie within the interval  $[-0.5, 0.5]$ . This is not mathematically necessary, but facilitates prior specification and interpretation. The prior distribution for each terminal node parameter is then given by

$$\mu_{jk} \sim \mathcal{N}(0, \sigma_\mu^2)$$

where

$$\sigma_\mu = \frac{0.5}{k\sqrt{m}}$$

$m$  is the number of trees,  $k$  is a shrinkage parameter. The constant 0.5 encourages the sum of the tree contributions to remain within a range compatible with the rescaled outcome. The prior aims to shrink the terminal node parameters toward zero, which leads each tree to contribute only a small portion of the variation in the outcome. Both a higher number of trees  $m$  and higher values of the shrinkage parameter  $k$  result in stronger shrinkage toward zero. The recommended default parameter by (Chipman et al., 2010) for  $k$  is 2.  $m$  is usually set to at least 50, while 100 or 200 are also common choices (Tan & Roy, 2019).

### 2.3.3 Error variance

Another prior is placed on the error variance  $\sigma^2$ . In the original BART specification,  $\sigma^2$  follows an inverse-chi-square distribution:

$$\sigma^2 \sim \frac{\nu\lambda}{\chi_\nu^2} \quad (2.7)$$

where  $\nu$  denotes the degrees of freedom and  $\lambda$  is a scale parameter.  $\nu$  controls the shape of the distribution and the heaviness of its tails. Smaller values of  $\nu$  result in heavier tails, allowing for more flexible variance estimates.  $\lambda$  determines the scale of the prior distribution and is not fixed directly. It is chosen in a data-driven way based on the empirical variance of the outcome variable,  $\hat{\sigma}_y^2$ , such that

$$P(\sigma < \hat{\sigma}_y) = q,$$

where  $q$  is the fraction of prior mass placed below the empirical standard deviation of the outcome  $\hat{\sigma}_y$ . ~~nu~~ and  $q$  are often specified as a pair. Typical values recommended by ~~Chipman et al., 2010~~ are the conservative specification (10, 0.75) which places more prior mass on smaller variances, the more aggressive specification (3, 0.99), which allows for much larger variances, and (3, 0.90), which avoids extremes and is often used by default.

## 2.4 Markov Chain Monte Carlo Algorithm

The posterior over the BART parameters (the tree structures  $T_j$ , the terminal node parameters  $M_j$ , and the error variance  $\sigma^2$ ) is estimated with a Markov Chain Monte Carlo (MCMC) algorithm. Sampling from the joint posterior of  $p((T_1, M_1), \dots, (T_m, M_m), \sigma^2 \mid Y, X)$  directly

is intractable. Therefore, a Gibbs sampler is implemented which iteratively samples from the full conditional distributions. Since it is not possible to sample directly from the conditional distribution of the tree structures, a Metropolis-Hastings step is embedded within the Gibbs sampler, resulting in a Metropolis-within-Gibbs algorithm. The terminal node parameters and the error variance can be drawn in closed form from their full conditional distributions.

The algorithm follows a Bayesian backfitting scheme, where each tree is updated conditional on all the other trees. Instead of fitting each tree to the outcome  $Y$ , each tree is fitted to the partial residuals, which represent the variation in the outcome not explained by the other trees.

One iteration of the Gibbs sampler proceeds as follows:

First, the partial residuals are calculated as

$$R_j = Y - \sum_{l \neq j} g(X; T_l, M_l)$$

Conditional on  $R_j$ , a new tree structure  $T_j^*$  is proposed by making a small random change to the current tree structure  $T_j$ . The options for modifying the tree typically are:

- grow: split one of the current terminal nodes,
- prune: remove an existing split,
- change: modify the splitting rule of an internal node,
- swap: exchange the splitting rules of two internal nodes.

The type of modification is randomly chosen. A proposal distribution assigns a probability to each of the modifications. Common implementations place higher probability on grow and prune moves than on change and swap moves. In case of a grow move the covariate and the splitting value are selected according to the priors defined in 2.3.1.

To decide whether to accept the new tree  $T_j^*$  or keep the current tree  $T_j$ , the Metropolis-Hastings acceptance ratio is calculated.

$$\alpha = \min \left( 1, \frac{p(R_j | T_j^*) p(T_j^*) q(T_j | T_j^*)}{p(R_j | T_j) p(T_j) q(T_j^* | T_j)} \right)$$

This ratio compares whether the proposed tree structure fits the data better than the old tree structure. It consists of three components.

The first component is the likelihood ratio

$$\frac{p(R_j | T_j^*)}{p(R_j | T_j)}$$

which measures whether the proposed tree improves the fit to the partial residuals compared to the current tree. The likelihood

$$p(R_j | T_j) = \int p(R_j | T_j, M_j) p(M_j | T_j) dM_j$$

is the marginal likelihood of the tree. The terminal node parameters  $M_j$  are integrated out analytically, which is possible because conjugate priors are used for the terminal node parameters. Using the marginal likelihood instead of conditioning on  $M_j$  improves the efficiency of the MCMC algorithm.

The second component

$$\frac{p(T_j^*)}{p(T_j)}$$

is the ratio of the priors of the two tree structures. It acts as a penalty for complexity, favoring shallow trees. Larger trees should only be accepted if they improve the fit enough to justify additional splits.

The third component is the proposal ratio

$$\frac{q(T_j | T_j^*)}{q(T_j^* | T_j)}$$

which adjusts for the fact that some modifications are more likely than others (e.g. a large tree offers more opportunities for pruning than a small one). This correction ensures that the Markov chain samples from the correct posterior distribution.

If the proposed tree is accepted,  $T_j$  is replaced by  $T_j^*$ . Otherwise  $T_j$  is kept.

Conditional on the updated tree structure, the terminal node parameters are sampled from their full conditional posterior distributions

$$p(\mu_{jk} | T_j, R_j)$$

which are normally distributed.

After all  $m$  trees have been updated, the error variance  $\sigma^2$  is sampled from its full conditional posterior distribution.

The above procedure describes one draw from the posterior distribution. At the beginning,

the trees are initialized consisting only of a root node. The sampling procedure is then repeated many times, disregarding a specified number of burn-in samples to reduce the influence of the starting values. After a sufficient number of iterations, the generated draws approximate samples from the posterior distribution. These posterior draws can then be used to generate predictions for new data and to construct Bayesian credible intervals.



## 3 Causality

### 3.1 Potential Outcomes Framework

To formally define causal effects throughout this thesis, the potential outcomes framework (POF) will be used. As it was introduced by Neyman (1990) (originally published 1923) and extended by Rubin (1974) to observational studies, it is also known as the Neyman-Rubin causal model (NRCM).

A population or sample consists of  $n$  units indexed by  $i = 1, \dots, n$ . For each unit,  $X_i$  denotes the vector of observed covariates,  $Y_i$  denotes the outcome variable, and  $Z_i$  denotes the treatment assignment, indicating whether the unit receives treatment or not. In the case of binary treatment assignment,  $Z_i = 1$  indicates that the unit receives treatment,  $Z_i = 0$  indicates that the unit does not. For each unit, there are two potential outcomes:  $Y_i(1)$  is the outcome that would be observed if the unit received treatment,  $Y_i(0)$  is the outcome that would be observed if the unit did not receive treatment (Rubin, 1974).

To illustrate the concepts, the effect of tutoring on students' test scores will be used as an example throughout this thesis. Here, the units are the students,  $Z_i$  indicates whether a student attends tutoring classes, and  $Y_i$  is the student's test score.  $Y_i(1)$  would be the test score of student  $i$  if they attended tutoring classes.  $Y_i(0)$  would be the test score if they did not.

Based on the potential outcomes, several estimands can be defined. The individual treatment effect (ITE) is the causal effect that the treatment has on an individual unit and is defined as the difference between the two potential outcomes of that unit.

$$\tau_i = Y_i(1) - Y_i(0)$$

Since the two potential outcomes refer to the same unit, they differ only in treatment status. Therefore, any difference observed between the potential outcomes can be attributed to the treatment.

This calculation is, however, purely conceptual, because in reality a unit either receives treatment or it does not. Therefore, only one of the outcomes can be observed, while the other remains unobserved and is referred to as the counterfactual outcome. This is known

### 3 Causality

as the fundamental problem of causal inference. Since both potential outcomes cannot be observed simultaneously for the same unit, the individual treatment effect cannot be directly identified. Because of that, causal inference typically focuses on aggregate measures of causal effects.

The average treatment effect (ATE) is the average causal effect of the treatment across the entire population:

$$ATE = \tau = E[Y(1) - Y(0)]$$

The sample average treatment effect (SATE) can be written as the average of the individual treatment effects:

$$SATE = \tau_S = \frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0))$$

Although the individual treatment effects are not directly observable, the ATE can be estimated through comparisons between treatment and control group under certain assumptions, which will be discussed in 3.5.

Treatment effects can differ across subpopulations. In the tutoring example, tutoring could work better for students from certain socio-economic backgrounds. When treatment effects are evaluated conditional on covariates, the conditional average treatment effect (CATE) is obtained:

$$CATE = \tau(x) = E[Y(1) - Y(0) \mid X = x]$$

If treatment effects vary across units depending on their covariates, the treatment effect is referred to as heterogeneous. If the treatment effect is identical for all units, it is referred to as homogeneous. In this case, the CATE is equal to the ATE for all covariate values.

Two additional important estimands are the average treatment effect on the treated (ATT), which measures the average causal effect among units that actually received treatment:

$$ATT = E[Y(1) - Y(0) \mid Z = 1]$$

and the average treatment effect on the controls (ATC):

$$ATC = E[Y(1) - Y(0) \mid Z = 0]$$

Similar to the individual treatment effect, average treatment effects cannot be directly observed. However, under certain assumptions, they can be estimated through comparisons between treatment and control group. In observational settings, differences between these

groups may not only be caused by the treatment itself, but also by additional variables that influence the outcome and treatment assignment. These variables are referred to as confounders.

## 3.2 Confounders

A confounder is a variable that has an influence on both the outcome and on treatment assignment (Pearl, 2009). This creates a problem because the observed association between treatment and outcome cannot be attributed solely to the treatment. Instead, some or even all of the observed association may be driven by the confounder, which affects both variables. As a result, estimates of treatment effects may be biased. The relationship between confounders, treatment and outcome is displayed in Figure 3.1.

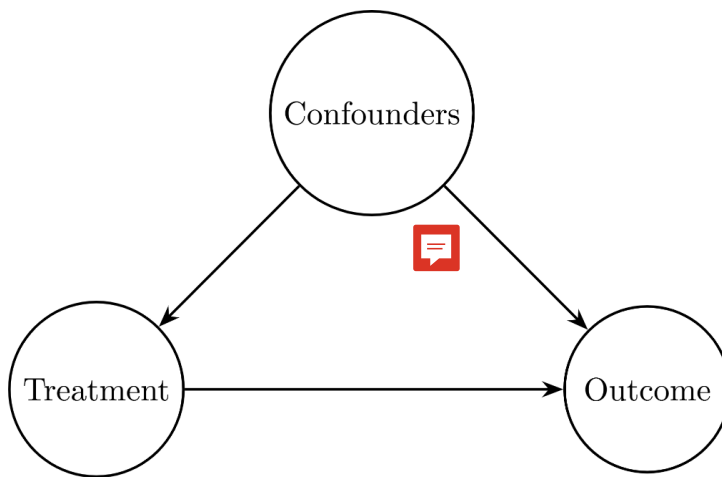


Abbildung 3.1: Relationship between confounders, treatment assignment and outcome

In the tutoring example, a confounder could be previous academic performance. Students who previously have lower grades, are typically more likely to attend tutoring classes. They are also more likely to get a lower test score. Previous grades therefore have a relationship with both the treatment and the outcome. Ignoring this confounder could lead to misleading conclusions about the effectiveness of tutoring. In this case, it could falsely suggest that attending tutoring classes leads to lower test scores, even if the actual causal effect is positive.

To still obtain valid estimates of causal effects, it is necessary to control for confounding variables (Pearl, 2009). The goal is to remove the part of the association between the treatment assignment and the outcome, that the confounders induce. Then, only the

effect of the treatment on the outcome remains. To identify relevant confounders, domain knowledge is often required (Chen et al., 2024).

### 3.3 Randomized controlled trials

The standard approach of estimating causal effects in scientific research are randomized controlled trials (RCTs) (Shadish et al., 2002). Here, participants are divided into two groups. One group, called the treatment or experimental group, receives the treatment, the other, called the control group, does not. An important aspect of RCTs is that the assignment to either treatment or control group happens completely at random. Since there are no variables that influence who is assigned to which group, the groups are on average comparable in terms of observed and unobserved variables. As a result, systematic differences in the outcome variable ~~can be~~ assumed to be caused by the treatment, not by other covariates (Stolberg et al., 2004). Apart from controlling for known confounders, randomization also controls for unknown confounders already at the design stage of the study.

While RCTs are considered the gold standard for causal inference, they are not always feasible. Conducting them can then can be expensive or impossible (Shadish et al., 2002). Therefore, observational studies are very common in practice.

### 3.4 Observational studies

Observational studies are studies where the assignment of the treatment is not under the control of the researcher (Imbens & Rubin, 2015). While RCTs are conducted with the explicit goal of investigating a specific research question, observational studies often use data that was collected for other purposes rather than for the study itself. This typically makes observational studies less costly and more flexible in scope (Shadish et al., 2002).

In the tutoring example, the data may not come from a controlled experiment but instead from a school database with student information. The database records test scores and whether a student attended tutoring. The decision to assign a student to tutoring was made by the parents or the teacher rather than through random assignment. Treatment assignment is therefore likely influenced by covariates such as prior academic performance or socio-economic background. This induces systematic differences between the treatment and control group, leading to confounding (Rosenbaum & Rubin, 1983). In contrast to RCTs, which control for confounders through randomization at the design stage, observational

studies must address confounding in the analysis stage (Imbens & Rubin, 2015).

## 3.5 Assumptions for Causal Inference

To estimate causal effects from non-randomized data, several assumptions are required.

### 3.5.1 Ignorability Assumption

The ignorability assumption, also referred to as unconfoundedness assumption, assumes that there are no unobserved confounders (Rubin, 1974). In order to obtain unbiased estimates of treatment effect, all relevant confounders must be observed and controlled for. The assumption can be formally written as (Rosenbaum & Rubin, 1983):

$$Y(0), Y(1) \perp Z \mid X$$

This means that, conditional on the observed covariates  $X$ , treatment assignment  $Z$  is independent of the potential outcomes. In other words, once controlled for  $X$ , there are no other factors that influence treatment assignment and outcome. This assumption cannot be tested directly, since by definition, unobserved confounders cannot be observed.

Previous grades as a possible confounder in the tutoring example, were already discussed in 3.2. If previous grades are not observed and controlled for, the ignorability assumption is violated and estimates may be biased.

### 3.5.2 Overlap Assumption

The overlap assumption, also referred to as common support assumption, requires that for every covariate pattern there is a positive probability of both receiving and not receiving treatment (Rosenbaum & Rubin, 1983):

$$0 < P(Z = 1 \mid X = x) < 1 \quad \text{for all } x$$

That means that there are no subgroups of individuals who always receive or who never receive treatment. The assumption is important because treatment effects can only be estimated when treated and untreated individuals with comparable covariate values exist.

In the tutoring example, the assumption would be violated if all students from a certain socio-economic background were assigned to tutoring and not a single person from this subgroup would remain in the control group.

#### 3.5.3 Stable Unit Treatment Value Assumption

The stable unit treatment value assumption (SUTVA) assumes that each unit has only one potential outcome under each treatment condition (Rubin, 1980). This assumption is required to ensure the stability and interpretability of causal effects.

Two conditions need to be fulfilled for SUTVA to hold:

1. There is no interference between units. This means that the potential outcomes of one unit are not affected by the treatment status of another unit. In the tutoring example, the assumption would be violated if students attending tutoring shared learning material with students in the control group, influencing the test scores of the control group.
2. ~~2~~ There are no different variations of treatment. All treated units must receive the same version or intensity of the treatment. In the tutoring example, this would require all students in the treatment group to attend the same number and type of tutoring sessions.

## 4 Causal Inference Methods

### 4.1 From Predictive to Causal Machine Learning

Standard machine learning models aim to predict outcomes accurately, causal inference on the other hand focuses on the estimation of treatment effects. Predictive associations do not necessarily imply causal relationships, therefore predictive machine learning methods cannot automatically be applied to causal tasks (Shmueli, 2010).

A variety of methods have been developed for the estimation of treatment effects (a review of current research trends can be found in Arti et al. (2020)). These methods have different approaches in addressing confounders, either by trying to create a population which mimics an RCT or by modelling the underlying data-generating processes. In general, there are two mechanisms that are of interest (?). One is the treatment assignment mechanism, which describes the process determining which individuals receive treatment and which do not, represented by  $T | X$ . The second is the outcome mechanism, which captures how the covariates and the treatment determine the outcome together, represented by  $Y | T, X$ . Many causal machine learning methods estimate these mechanisms using so-called nuisance models (Chernozhukov et al., 2018). These are models which are not of primary interest, but are used to control for confounding and improve treatment effect estimation.

In this chapter, a short introduction to the methods used throughout this thesis will be provided.

#### 4.1.1 Implications for Model Evaluation

Due to the differences between predictive and causal models, two things need to be considered when evaluating causal inference methods.

##### Full-Sample Estimation

Predictive models aim to predict the outcome of new observations well. A good model should not only learn the relationship in the data it is trained on, but be able to generalize well to unseen data. Therefore, when evaluating predictive models, data is typically split

into training and test data (Stone, 1974). The model is fit to the training data and evaluated on the test data.

In contrast, in causal inference the goal is not primarily outcome prediction, but the estimation of causal parameters, mainly the treatment effect. This task is complicated by the fact that the ground truth is never observed (Athey & Imbens, 2016). Because standard out-of-sample validation cannot be applied in this setting, causal effect estimation is often performed using the full dataset in order to make use of all available information.

### Data Sources for Evaluating Causal Effect Estimators

The difference in evaluation strategy extends further: because standard predictive validation cannot confirm whether a causal estimate is correct, researchers additionally face the challenge of how to evaluate causal estimators at all. The best option seems to be real-world data, as it ensures that the results are relevant for practical applications. However, this leads to the fundamental problem of causal inference: The true treatment effect is not known, since for each unit, only one potential outcome can be observed at a time (Holland, 1986). As a result, model performance cannot be evaluated directly.

For this reason, synthetic data is often used. In this case, the covariates, the treatment assignment mechanism and the outcome mechanism are simulated. The ground truth is known and model estimates can be directly compared to true treatment effects. Additionally, key characteristics of the data, such as overlap or the degree of treatment effect heterogeneity, can be systemically varied.

Synthetic data has some clear limitations: It is often less complex than real-world data, therefore results from simulation studies may not transfer well to practical problems (Curth et al., 2021). Another issue with simulated data is that researchers can choose data-generating processes which favor some methods over others, especially if they align with the assumptions of those methods.

To address these issues, semi-synthetic data is often favored over synthetic data (Neal et al., 2020). Here, the covariates and in some cases the treatment assignment come from real-world data. Only the outcome model and sometimes treatment assignment is simulated. This introduces some of the complexity of real data, while still allowing for a known ground-truth.

## 4.2 Propensity Scores

A widely used tool in causal inference is the propensity score. The propensity score is the probability of receiving treatment conditional on the covariates (Rosenbaum & Rubin,



1983):

$$e(X) = P(Z = 1 \mid X)$$

It captures the relationship between the covariates and treatment assignment and can therefore be used to control for confounding. An advantage of propensity scores is that they can reduce a high-dimensional set of covariates to a single variable (Olmos & Govindasamy, 2015).

Propensity scores are typically estimated using logistic regression or other binary classification models (Rosenbaum & Rubin, 1983; Lee et al., 2010). While propensity scores are not a standalone method for treatment effect estimation, they are often incorporated into other methods. An example is propensity score matching, where each observation from the treatment group is matched to one or more observations from the control group with similar propensity scores and only those observations are included in further analysis models (Rosenbaum & Rubin, 1983). This way, differences in covariate distributions between treatment and control group, which are introduced by confounders, are reduced.

### 4.3 Meta-learners

A common approach to estimating heterogeneous treatment effects is to adapt standard predictive machine learning models using so-called meta-learners. A meta-learner is a framework in which one or more predictive models can be embedded (Künzel et al., 2019). It defines a strategy for how these models can be used to estimate treatment effects.

The S-learner (single learner) uses a single model, in which the treatment indicator is included as an additional covariate (Künzel et al., 2019):

$$\hat{\mu}(x, w) = E[Y \mid X = x, Z = z] \quad (4.1)$$

For estimating the conditional average treatment effect for a unit, predictions are made twice for each observation. While keeping all other covariates fixed, the treatment indicator is set once to  $Z = 1$  and once to  $Z = 0$ . The difference between these predictions is the estimate of the conditional average treatment effect:

$$\hat{\tau}(x) = \hat{\mu}(x, 1) - \hat{\mu}(x, 0) \quad (4.2)$$

The T-learner (two learners) fits two models separately, one using the observations in the treatment group and one using the observations in the control group (Künzel et al., 2019):

$$\begin{aligned}\hat{\mu}_1(x) &= E[Y \mid X = x, Z = 1] \\ \hat{\mu}_0(x) &= E[Y \mid X = x, Z = 0]\end{aligned}\tag{4.3}$$

Each model then generates predictions for all observations, their difference estimates the conditional average treatment effect:

$$\hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x)\tag{4.4}$$

S- and T-learners both are fitted using the observed outcomes. For the estimation of treatment effects, they predict both the factual and the counterfactual outcome.

There exist other meta-learners like X-learners or R-learners (Künzel et al., 2019; Nie & Wager, 2021). For the sake of brevity, this thesis focuses only on S- and T-learners.

### 4.3.1 Strengths and limitations of meta-learners

A limitation of the T-Learner is that it can only use a subset of the data for each of its models, which can be especially problematic if the dataset is small or imbalanced (when one group contains substantially more observations than the other) (Künzel et al., 2019). In these settings, the S-learner has an advantage, since it uses information from all observations for a single model. This is particularly beneficial when treatment effects are relatively homogeneous.

The S-learner may however struggle when the conditional outcome functions  $\mu_1(x)$  and  $\mu_0(x)$  differ substantially, i.e. when treatment effects are highly heterogeneous (Zhang et al., 2025). The T-Learner can better account for differences between the groups because it fits two separate models.

### 4.3.2 Effects of Regularization

Two further limitations arise for S-learners when they are used in combination with regularized predictive models. In high-dimensional settings, the inclusion of many covariates can reduce bias but may also increase variance (Tibshirani, 1996). Therefore, regularization is used to only include covariates in the model that have strong predictive power, i.e. by penalizing a high number of covariates. Since S-learners treat treatment assignment like any other covariate, regularization may shrink the treatment effect towards zero or exclude treatment assignment from the model (Künzel et al., 2019).

A regularized model may also weight down or exclude confounders that have a weak relationship with the outcome but a strong relationship with treatment assignment. As a

result, the confounding is not sufficiently controlled for, leading to biased treatment effect estimates. This phenomenon is referred to as regularization-induced confounding (Hahn et al., 2018).

## 4.4 BART for Causal Inference

As BART showed strong predictive performance due to its flexible non-parametric structure, J. L. Hill (2011) proposed its use for causal inference. In this original approach, BART is applied as an S-learner. The sum-of-trees model can then be written as

$$Y = f(X, Z) + \varepsilon \quad (4.5)$$

with

$$f(X, Z) = \sum_{j=1}^K g(X, Z; T_j, M_j) \quad (4.6)$$

where  $Z$  denotes treatment assignment and each function  $g(\cdot)$  represents a single regression tree with tree structure  $T_j$  and corresponding terminal node parameters  $M_j$ .

Later work by Künzel et al. (2019) implemented BART as a T-learner. Hahn et al. (2020) introduced another variation to address the issue of regularization-induced confounding. In BART, this issue can arise because the priors encourage shallow trees and shrink the terminal node parameters toward zero. To address this, they recommended including an estimate of the propensity score as an additional covariate in the model, resulting in

$$Y = f(X, Z, \hat{e}(X)) + \epsilon, \quad (4.7)$$

where  $\hat{e}(X)$  denotes the estimated propensity score.

This way, information about treatment assignment is explicitly included in the model. The propensity score can be estimated using different methods, such as logistic regression or BART itself. Hahn et al. (2020) referred to this extension as ps-BART.

Since its introduction, the basic causal BART model has been extended to a wide range of settings, among these are continuous or multi-categorical treatments (Souto & Louzada Neto, 2024) or time-to-event outcomes (Kabata et al., 2026). These extensions are beyond the scope of this thesis, as they are not part of the empirical evaluation.

## 4.5 Benchmark methods

In addition to BART, this thesis considers two established benchmark methods for estimating treatment effects: augmented inverse probability weighting (AIPW) and causal forests. Unlike causal BART, neither method is based on the meta-learner framework. Instead, both target the causal estimand directly.

### 4.5.1 Augmented Inverse Probability Weighting

AIPW combines a model for treatment assignment with a model for the outcome mechanism to improve robustness against model misspecification, an idea originating in semiparametric missing-data theory (Robins et al., 1994) and later adapted to causal inference (Bang & Robins, 2005). Its treatment assignment component is based on inverse probability of treatment weighting (IPTW) (Robins et al., 2000), which aims to mimic random treatment assignment by creating a pseudo-population in which treatment is independent of the observed confounders. This is achieved by estimating propensity scores for each observation and weighting the observations with the inverse probability of the treatment they actually received.

For treated observations, Robins et al. (2000) define the weights as

$$w_i = \frac{1}{e(X_i)}$$

and for control observations as

$$w_i = \frac{1}{1 - e(X_i)}$$

This way, observations that have a low probability of being in their respective group receive a higher weight and those who have a high probability are assigned a lower weight. As a result, overrepresented observations are downweighted and underrepresented observations are upweighted, which leads to more balanced covariate distributions between the groups (Robins et al., 2000; Chesnaye et al., 2022).

When propensity scores are close to 0 or 1, the weights become extreme (Cole & Hernán, 2008). This can result in a few observations influencing the result disproportionately and increased variance. To reduce this problem, stabilized weights are often used, where the numerator 1 in the weighting formulas is replaced exchanged with the marginal treatment probability, which is done by exchanging the numerator 1 in formula X to the marginal

treatment probability, which is the proportion of treated or untreated units in the full sample.

Since IPTW uses propensity scores as weights, results can vary depending on how well the propensity score model captures the true treatment assignment model. To increase robustness against model misspecification, augmented inverse probability weighting (AIPW) additionally incorporates a model for the outcome mechanism (Robins et al., 1994). Instead of directly using the outcome model to predict the treatment effect, the weighted outcomes are included in the predictions through an augmentation term. The estimator of the ATE can be written as

$$\hat{\tau}_{AIPW} = \frac{1}{n} \sum_{i=1}^n \left[ \frac{Z_i Y_i - (Z_i - \hat{e}(X_i)) \hat{\mu}_1(X_i)}{\hat{e}(X_i)} - \frac{(1 - Z_i) Y_i + (Z_i - \hat{e}(X_i)) \hat{\mu}_0(X_i)}{1 - \hat{e}(X_i)} \right],$$

where  $\hat{\mu}_1(X_i)$  and  $\hat{\mu}_0(X_i)$  denote the predicted outcomes under treatment and control from the outcome model. Methods that combine a treatment assignment model and an outcome model this way are referred to as “doubly robust” (Scharfstein et al., 1999). Their strength lies in that even if one of the models is misspecified, they can still produce consistent estimates.

Weighting causes some observations to contribute more strongly to the pseudo-population than others (Robins et al., 2000). Therefore, when calculating standard errors, the weighting procedure must be taken into account. Typically bootstrap methods are used to estimate standard errors.

#### 4.5.2 Causal forest

Causal forests are an ensemble tree ensemble approach for causal effect estimation. All descriptions in this section are based on the paper from Wager & Athey (2018), who developed the method. As discussed in 2.1, in regression trees used for prediction, splitting rules are chosen so that observations within the same terminal node have similar outcomes. In contrast, causal forests aim to detect heterogeneity in treatment effects. Splits are selected to maximize differences in treatment effects between terminal nodes. The goal is to separate observations with different treatment effects and to group observations with similar treatment effects together. Within each terminal node, treatment effects are estimated by comparing the outcomes of treated and control observations.

Randomness is introduced to avoid all trees in the ensemble having the same structure. Each tree is built on only a random subsample of the data and at each internal node, only a

random subset of the covariates is considered for determining the splitting rule. Additionally, causal forests typically estimate nuisance models for the treatment assignment mechanism and the outcome mechanism. These models are used to remove variation explained by the covariates before estimating treatment effect heterogeneity. This process is referred to as local centering.

To reduce overfitting, causal forests commonly employ so-called honest trees. The data is split into two parts, one subsample is used for determining the tree structure, the other is used to estimate the treatment effects within the terminal nodes. This separation reduces bias in treatment effect estimation.

### Hyperparameters

Wager & Athey (2018) also introduced several hyperparameters which can be tuned. First, there is the number of trees in the forest. A larger number of trees generally reduces the variance of the estimator, however, with more trees computational costs increase.

Another important hyperparameter is the number of covariates that are considered when determining the splitting rule at an internal node. Lower values reduce correlation between trees, but considering too few covariates may result in important covariates being excluded from the splitting process too often.

The minimum node size determines how many observations a terminal node must at least contain, therefore controlling when splitting stops. Lower values allow the causal forest to detect more heterogeneous treatment effects, but very small terminal nodes can increase the risk of overfitting.

Another tunable parameter is the subsample fraction, i.e. the proportion of observations used to build each tree. Larger subsamples may reduce variance but can increase similarity between the trees.

Additional hyperparameters include the proportion of data used for splitting rules versus estimation under honesty, whether empty terminal nodes should be pruned, the maximum allowed imbalance of a split and penalties for imbalanced splits. These parameters mainly affect the bias-variance trade-off and the stability of treatment effect estimation.

## 4.6 Empirical Performance of BART and Benchmark Methods

In general, causal BART models have shown strong empirical performance across a wide range of studies and datasets. They often outperformed AIPW estimators (McConnell & Lindner, 2019; Kim et al., 2023) as well as causal forests and other tree-based methods in

terms of bias and coverage (Wendling et al., 2018; Hahn et al., 2020; Miratrix et al., 2025; Caron et al., 2022; Lu et al., 2018).

When comparing different meta-learners in combination with BART, results are mixed and appear to be dependent on the underlying data-generating process. Some studies found that S-learners achieve lower RMSE than T-learners across a wide range of data settings (Hahn et al. (2020); Miratrix et al. (2025)). Other studies report advantages for the T-learner in settings with more complex treatment effect heterogeneity. For example, (Caron et al., 2022) found that T-learners outperformed S-learners on a dataset, where the outcome functions for treatment and control group differed substantially. In contrast, S-Learners performed better in a second simulation setting with less complex treatment effect functions. Similarly, Künzel et al. (2019) found that S-learners performed better than T-learners in small, imbalanced samples, however, they used a random forest as base learner instead of BART. Overall, the literature suggests that no single meta-learner consistently dominates across all settings.

Regarding the inclusion of propensity scores, Hahn et al. (2020) found that PS-BART achieved lower bias and improved coverage compared to standard BART across several simulated scenarios and the ACIC 2016 dataset, although the improvements were smaller than the differences between S- and T-Learners. McConnell & Lindner (2019) similarly reported lower bias for PS-BART when propensity scores were estimated using logistic regression. In contrast, Wendling et al. (2018) found only modest and inconsistent improvements of PS-BART over standard BART in simulated settings for heterogeneous treatment effect estimation.





## 5 Methodology

### 5.1 Research Goal

Despite the general strong performance of BART models for causal effect estimation, results are not always clear. This is especially true, when it comes to the comparison of different BART learners to each other. Additionally, hyperparameter tuning is usually neglected completely. Therefore, the goal of this thesis is to provide a systematic comparison of causal BART learners, both with each other and with alternative methods. Three research questions will be evaluated:

1. How does the performance of different causal BART learners compare?
2. How do BART-based approaches compare to other causal effect estimation methods?
3. To what extent can hyperparameter tuning improve model performance, and can tuning alter which method performs best?

### 5.2 Data Source

As discussed in ?? real-world data does not allow for the evaluation of causal estimators against a known ground truth. In this thesis, the objective is to compare models in terms of how well they estimate treatment effects. This requires data for which the true treatment effect is known. At the same time, simulating data is avoided in order not to impose data-generating processes which may favor certain models. On the other hand, it is important to compare model performance across different data settings to be able to assess robustness under varying circumstances. Therefore, (semi-)synthetic data generated by external sources is used, ensuring that the underlying data-generating processes are not specified within this thesis. Among available benchmark datasets, the choice fell on data from the Atlantic Causal Inference Conference (ACIC) 2019 Data Challenge (Gruber et al., 2019) as it provides a diverse collection of datasets generated under different data-generating processes.

The ACIC Data Challenge is a yearly competition held at the Atlantic Causal Inference Conference since 2026, which focuses on solving challenges in causal inference (Dorie et al.,

2019). Datasets containing covariates, treatment assignments and outcomes are provided by the organizers and participants are tasked with estimating different causal parameters. The participants can submit models of their choice to solve the tasks. After the challenge closes, the true treatment effects are disclosed and the winning models are announced.

The datasets have become popular beyond of the data challenge and are often used as benchmark datasets in causal inference research (e.g. (Mahajan et al., 2022; Ramadhanti et al., 2025; Wang et al., 2026)). They offer the opportunity to evaluate methods on a range of datasets generated independently of the researchers.

### 5.3 Dataset Characteristics

In the 2019 challenge, the task was to provide point estimates of the ATE as well as 95% confidence intervals. 64 different DGPs were provided, with 100 datasets generated for each DGP. All datasets featured binary treatment assignment. There were two different tracks, one for low-dimensional datasets and one for high-dimensional datasets. Within each track, both binary and continuous outcomes were included.

Tabelle 5.1: Characteristics of the data-generating processes

| DGP | Number of Covariates | Number of Observations | True ATE |
|-----|----------------------|------------------------|----------|
| 29  | 26                   | 5735                   | 4        |
| 30  | 26                   | 5735                   | 10       |
| 31  | 26                   | 5735                   | 2        |
| 32  | 26                   | 5735                   | 2.527    |
| 37  | 22                   | 500                    | 4        |
| 38  | 22                   | 500                    | 2.500    |
| 39  | 22                   | 500                    | 6.270    |
| 40  | 22                   | 500                    | 6.962    |
| 57  | 31                   | 649                    | 0.500    |
| 58  | 31                   | 649                    | 1        |
| 59  | 31                   | 649                    | -0.800   |
| 60  | 31                   | 649                    | 1.686    |
| 61  | 29                   | 649                    | 2        |
| 62  | 29                   | 649                    | 2        |
| 63  | 29                   | 649                    | -2       |
| 64  | 29                   | 649                    | 3.625    |

For this thesis, only the low-dimensional DGPs with continuous outcomes were selected,

resulting in 16 DGPs or 1600 datasets. Table 5.1 provides an overview over the selected datasets. The original names of the DGPs are retained, therefore DGP identifiers are not consecutive. The number of covariates ranges from 22 to 31, the number of observations from 500 to 5735.

For some DGPs, all covariates were simulated, for others the covariates were taken from real-world data sets. In all cases, the treatment assignment mechanism and outcome mechanism were simulated. The assumption of unconfoundedness was satisfied for all data sets. The organizers stated that various difficulties were incorporated into the datasets, such as non-linearity of ... (TODO direct quote). It was not disclosed which specific characteristics were present in each DGP. This makes the datasets suitable for evaluating the robustness of causal inference methods across a range of unknown data-generating conditions.

## 5.4 Data Diagnostics



The two measures calculated to characterize the data were balance and overlap. Both measures were calculated across 5 datasets per DGP and results were averaged.

The balance can be directly calculated as the proportion of observations in the treatment group in relation to the total number of observations.

The overlap is the degree to which treatment and control group share similar covariate distributions (TODO: reference to overlap assumption). Exactly quantifying overlap is not straightforward in high-dimensional settings. Common estimation approaches include comparing the propensity score distributions between treatment and control groups or identifying the proportion of observations with extreme propensity scores (Austin, 2011). These methods rely on correct propensity score estimation and do not account for the fact that not all covariates contribute equally to the outcome mechanism (J. Hill & Su, 2013). A method based on propensity scores might identify insufficient overlap for a covariate that is irrelevant for the outcome and is therefore not a genuine confounder.

J. Hill & Su (2013) therefore proposed an alternative approach based on the posterior draws of BART. The underlying idea is that in regions with good overlap, the model has enough information from both the treatment and control group and can make stable estimates of individual treatment effects. In regions with poor overlap, the model must extrapolate to generate counterfactual predictions, which leads to increased uncertainty of those predictions. Therefore, observations that do not have a counterfactual in the same covariate region can be identified by comparing the variance of their predicted counterfactual outcomes to the variance of the factual outcome predictions. If the variance of the counterfactual predictions

is substantially higher, this serves as an indicator that there is no adequate counterfactual in that covariate region.

The method was used to estimate the level of overlap in the datasets. Based on a rule defined by J. Hill & Su (2013), observations were flagged if their counterfactual outcome standard deviation  $s_i^{1-a}$  exceeded the maximum factual standard deviation across all observations in the same treatment group, plus one standard deviation of that factual distribution:

$$s_i^{1-a} > m_a + \text{sd}(s_j^a)$$

where  $m_a = \max_j \{s_j^a\}$  for all  $j$  with  $Z_j = a$ , and  $\text{sd}(s_j^a)$  denotes the standard deviation of the factual outcome standard deviations across all units in the same treatment group.

### 5.5 Experimental Procedure

Four different BART meta-learners were compared to assess whether their methodological differences lead to differences in estimation performance. The four meta-learners were an S-learner, a T-learner and two S-learners which include estimates of the propensity scores. One estimated propensity scores using a logistic regression and is referred to as PS-GLM, the other estimated the propensity scores using a BART model and is referred to as PS-BART.

To place the performance of BART in context, the BART models were also compared with two established causal inference methods. First, AIPW was included as a parametric, doubly robust benchmark. Second, causal forests were included as a widely used non-parametric method which, similar to BART, is based on trees.

The empirical evaluation consisted of two stages. Both stages were repeated across all DGPs, allowing an assessment of how performance varies under different data conditions.

In the first stage, all models were estimated using the default hyperparameters provided by the respective software packages. This allowed a first evaluation of the performance differences. Furthermore, this may likely be a setting which practitioners use often.

In the second stage, hyperparameters were tuned on a subset of the DGPs for each model. This way, it could be investigated whether model performance can be improved under optimal hyperparameter settings and whether the ranking of methods can be changed by tuning.

## 5.6 Evaluation Metrics

To evaluate the performance of the models, several metrics are computed for each data generating process.

The root mean squared error (RMSE) measures the estimation accuracy of the ATE.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{\tau}_i - \tau)^2},$$

where  $\hat{\tau}_i$  is the estimated ATE of iteration  $i$ ,  $\tau$  is the true ATE of the respective DGP and  $n$  is the number of iterations. Lower RMSE values indicate greater estimation accuracy.

To assess uncertainty, at each iteration 95% uncertainty intervals are constructed and it is recorded whether the interval contains the true ATE. The interpretation of these intervals differs between frequentist and Bayesian methods. For the frequentist approaches causal forests and AIPW, confidence intervals are reported. If the ATE estimation were repeated and confidence intervals were constructed many times, 95% of these confidence intervals would be expected to contain the true ATE. In contrast, Bayesian methods, such as BART, construct credible intervals, which allow the statement that the true ATE lies within the credible interval with a posterior probability of 95%, conditional on the observed data. For each model and DGP, the average interval length and the coverage rate, defined as the proportion of intervals containing the true ATE, are calculated.

Since the DGPs differ in the magnitude of the true ATE, RMSE and confidence interval length are not directly comparable across DGPs. When the true ATE is large, DGPs may exhibit a disproportionately large RMSEs or interval lengths. This does not affect comparisons between models within a single DGP but can distort results when averaging across all DGPs. To address this issue, RMSE and average interval length were normalized by the true ATE of the respective DGP. Coverage is not affected by scaling and was therefore not normalized.

Not only the average performance across DGPs is of interest, but also the relative performance of the models. In practical applications, researchers usually select a single method. Therefore, it was evaluated how often each model performed best. The models were ranked within each DGP for each metric and summarized statistics were reported across all DGPs.

## 5.7 Model Implementation

All analyses were conducted in R (Quelle, Version checken).

The BART T-learner was implemented using the *dbarts* package (Quelle, version checkne). This package uses discrete trees, meaning the covariate values for the splitting rules are selected from a discrete set of covariate values, not a continuous one. The *bart2* function from the package was used because it allows for the specification of hyperparameters. Two separate BART models were fitted, one to the treatment group and one to the control group. Predictions were then made on both models for all observations. ITE estimates were calculated as the difference between the predicted treated and control outcomes.

The BART S-learner, PS-BART and PS-GLM were implemented using the *bartc* function from the *bartCause* package (Quelle, version checken). The package internally uses *dbarts*, so differences in performance cannot be attributed to different implementations. For all three models, a single model was fitted using the covariates and treatment assignment as input. The propensity score variants additionally included estimated propensity score estimates as covariates. Propensity scores were estimated within the *bartc* function. For PS-GLM, they were estimated using logistic regression, for PS-BART, a separate BART model was fitted with treatment assignment as outcome. ITE estimates were obtained by predicting the potential outcomes under treatment and control and calculating their difference.

For all BART models, the number of chains was set to 10, with 500 burn-in samples and 500 posterior samples per chain. These settings correspond to the default values of the *bartCause* package and are consistent with recommendations by Dorie et al. (2019) and Carnegie (2019). Convergence of the chains was assessed [LS13.1] using visual inspection of trace plots for the  $\sigma$  parameter.

Causal forests were implemented using the *causal\_forest* function from the *grf* package (Quelle, TODO version).

For AIPW, the augmented inverse probability weighting estimator was implemented using the AIPW package (Quelle, Version: 0.6.9.3). Both the outcome regression and the propensity score model were estimated via the SuperLearner package (Quelle, Version: 2.0.40) using a logistic model for treatment assignment and a linear regression for the outcome. Propensity scores were truncated at the 0.1 bound to avoid extreme weights.

### 5.8 Default Hyperparameters

The default hyperparameters of the *dbarts* package were used for all BART models:  $k = 2$ ,  $m = 75$ ,  $\alpha = 0.95$ ,  $\beta = 2$ ,  $(\nu, q) = (3, 0.90)$ .

For the causal forest, the default hyperparameters from the *grf* package were *num.trees* = 2000, *min.node.size* = 5, *sample.fraction* = 0.5, *honesty.fraction* = 0.5, *honesty.prune.leaves* =

$TRUE$ ,  $\alpha = 0.05$ , and  $imbalance.penalty = 0$ .  $mtry$  (number of variables considered at each split) was set dynamically within the package, as  $\sqrt{p} + 20$ , where  $p$  denotes the number of covariates.

## 5.9 Hyperparameter Tuning

BART models and causal forests both contain a number of hyperparameters. Due to limited computational resources, not all hyperparameters were tuned. Instead, only the parameters, for which literature expects a substantial impact on results, were tuned, while the others were kept at their default values. Hyperparameter tuning was conducted using a grid search.

Tuning of the BART models was based on the tuning in Chipman et al. (2010) and focused on the parameters governing the prior distribution of the terminal node values ( $k$  and  $m$ ) and the error variance ( $nu$  and  $q$ ). For  $k$ , the values (1, 2, 3, 5) and for  $m$  the value (75, 100, 200) were evaluated. For the variance prior, the combinations (3, 0.90), (3, 0.99), and (10, 0.75) were considered. This resulted in a  $4 \times 3 \times 3$  grid with a total of 36 combinations.

For the causal forest, tuning was based on Dandl et al. (2024) and focused on the parameters  $mtry$  (the number of covariates that are considered for choosing the splitting rule),  $min.node.size$  (minimum number of observations in a terminal node) and  $tuning.sample$  (proportion of observations used to build each tree). For  $min.node.size$ , values were (TODO), for  $mtry$  ( ) and for  $sample.fraction$  (0.3, 0.5). Higher values are not possible as at least half of the sample is needed for calculation of standard errors. This resulted in a  $4 \times 3 \times 2$  grid with a total of 24 combinations.

AIPW does not contain hyperparameters in the classical sense. However, both the propensity scores can be estimated through different models. A larger number of possible model specifications exists, including interaction terms and higher-order polynomials. Exploring all these possible models was beyond the scope of this thesis. Therefore, an alternative, more flexible method for propensity score estimation was evaluated. Instead of logistic regression, a BART model was used to model the treatment assignment mechanism.

For hyperparameter tuning, four DGPs were selected based on the results from the first stage of the experiment. The chosen DGPs had relatively big performance differences between the methods, so that potential improvements due to hyperparameter tuning could have a big impact. Additionally, they covered the full range of dataset sizes present in the study, both regarding the number of observations and the number of covariates. To reduce computational costs, only the first 20 datasets were considered for tuning. For each

## 5 Methodology

dataset, all methods were run with all the specified hyperparameter combinations. For each combination, the RMSE was calculated. The hyperparameter combination with the lowest RMSE was chosen for each method. Then, the RMSE from the tuned method and the RMSE from the default parameters were compared. For the default parameters, only the 20 first observations were considered.



## 6 Results

### 6.1 Data Diagnostics

Balance: DGPs 37 and 38 have the highest levels of imbalance ( $x \% x$ ), which resulted in a smaller treatment group with  $X$  respective  $X$  observations. DGP 40 was imbalanced as well ( $X \%$ ) but in the opposite direction, with a control group of  $X$  observations. All other DGPs had groups of at least 30 % (?) each. (TODO!)

Overlap: The ?HillSu2013 method identified three DGPs with poor overlap: DGP 39 (83.4 observations on average, 16.7 %), DGP 40 (15 observations, 3 %), and DGP 37 (10.4 observations, 2.1 %). Observations without adequate counterfactuals were identified in other datasets, but they did not make up a significant proportion of the dataset.

TODO: Table

### 6.2 Default Hyperparameters

#### 6.2.1 Aggregated Results

The following results provide an aggregated overview of the performance of the methods across all DGPs. Table 6.1 shows the mean normalized RMSE, normalized interval length and the coverage of each method, rounded to the third decimal. Figures (TODO) show boxplots of the metrics to provide an additional information on how the metrics are distributed across the DGPs. Note that the horizontal line in each boxplot marks the median, so it may differ from the mean values.

The S-learner had the lowest mean RMSE (0.091), closely followed by PS-GLM (0.092) and PS-BART (0.093). The mean RMSE of the T-learner was somewhat higher (0.099), while the causal forest reached 0.117 and AIPW has the highest RMSE of 0.171.

For coverage, the three S-learner-based BART models performed best again. Note that, in contrast to RMSE, higher coverage values are preferred. PS-BART had the highest coverage (0.939), followed by PS-GLM (0.936) and the S-Learner (0.933). The causal forest had a coverage of 0.913, the T-Learner of 0.893 and AIPW had the lowest coverage of 0.858.

Interval length roughly followed a similar pattern to RMSE. The S-Learner produced the

shortest intervals (0.357), followed by PS-GLM (0.363), the T-learner (0.366) and PS-BART (0.367). AIPW (0.384) and particularly, the causal forest (0.415) had the longest intervals.

Standard deviations across the 16 DGPs show that AIPW was the least stable model, with the highest variability in both RMSE ( $SD = 0.278$ ) and coverage ( $SD = 0.236$ ). The T-learner had the lowest RMSE variability ( $SD = 0.043$ ) but the second-highest coverage variability ( $SD = 0.102$ ). The three BART S-learners and the causal forest showed the most stable coverage (SDs between 0.046 and 0.076), while interval length variability was similar across all models (SDs 0.198–0.210).

Tabelle 6.1: Mean RMSE, coverage and interval length by model

| Model          | RMSE  | Coverage | Interval Length |
|----------------|-------|----------|-----------------|
| BART/PS-BART   | 0.093 | 0.939    | 0.367           |
| BART/PS-GLM    | 0.092 | 0.936    | 0.363           |
| BART/S-learner | 0.091 | 0.933    | 0.357           |
| BART/T-learner | 0.099 | 0.893    | 0.366           |
| Causal forest  | 0.117 | 0.913    | 0.415           |
| AIPW           | 0.171 | 0.858    | 0.366           |

Tabelle 6.2: Mean (SD) of RMSE, coverage and interval length by model, across all 16 DGPs

| Model          | RMSE          | Coverage      | Interval Length |
|----------------|---------------|---------------|-----------------|
| BART/S-learner | 0.091 (0.047) | 0.933 (0.046) | 0.357 (0.202)   |
| BART/PS-GLM    | 0.092 (0.048) | 0.936 (0.047) | 0.363 (0.208)   |
| BART/PS-BART   | 0.093 (0.050) | 0.939 (0.050) | 0.367 (0.210)   |
| BART/T-learner | 0.099 (0.043) | 0.893 (0.102) | 0.366 (0.202)   |
| Causal forest  | 0.117 (0.052) | 0.913 (0.076) | 0.415 (0.198)   |
| AIPW           | 0.171 (0.278) | 0.858 (0.236) | 0.366 (0.201)   |

### 6.3 Performance Across Data-Generating Processes

The metrics vary heavily between models as well as between DGPs. Figures X, Y and Z respectively show the normalized RMSE, coverage and normalized interval length for each method across all 16 DGPs.

### 6.3.1 Notable DGPs

Some DGPs stand out for having especially poor performance across multiple or all models.

DGP 30 had the highest average RMSE (0.259) and the lowest average coverage rate (0.648) across all models. It also produced the most extreme results of the whole study, with an RMSE of 1.196 and a coverage rate of 0.00 for AIWP. The T-Learner reaches 0.60, the causal forest 0.65. The S-Learner has a coverage of 0.84, PS-GLM of 0.86. Only PS-BART reached an acceptable coverage of 0.94.

DGP 39 had the lowest coverage for each of the three S-learner models (0.83 for the S-learner, 0.81 for PS-GLM and 0.77 for PS-BART) and a comparatively low coverage of the T-Learner (0.83).

DGP 31 had the second highest average RMSE (0.209) and AIWP (0.219) and the causal forest (0.248) performed substantially worse than the BART models (0.193 - 0.202).

DGP 57 had high RMSE values ( $\geq 0.163$ ) for all models. Note that it has a true ATE of only 0.5, so normalizing by such a small value will typically inflate the RMSE.

### 6.3.2 Methods evaluated across different DGPs

While the previous section highlighted DGPs with particularly notable results, this section systematically examines the performance of each method across all 16 DGPs.

#### BART S-learners

Across all DGPs, the RMSE of the three S-learners variants (S-learner, PS-GLM, PS-BART) was almost always very similar. Therefore, they are discussed together. The biggest difference between the three occurred at DGP 30, where the RMSE of PS-BART (0.049) was substantially lower than that of the S-learner and PS-GLM (0.058 each).

One of the three S-learners placed first in half of the DGPs (8 of 16). Independently, the S-learner was the best model in 3 DGPs (31, 38, 61) and PS-BART in 1 (30). In the remaining 4 DGPs, first place was shared: all three S-learner variants tied at DGPs 40 and 62, the S-learner tied with the T-learner at DGP 32, and with AIPW at DGP 39.

Even when outperformed by other models, the margin was typically small, e.g. for DGP64, the causal forest outperformed the best S-learner by only 0.002. The biggest difference between the worst S-learner and the best overall model is 0.014 between PS-BART and the T-learner at DGP 63. None of the S-Learners ranked last in any DGP.

When comparing the three S-learner variants only to each other, the S-learner was the sole best performer in 7 DGPs, PS-BART in 3, and PS-GLM never had the lowest RMSE

on its own. There were several ties: all three models tied at DGPs 40 and 62, PS-GLM and PS-BART tied at DGP 60, and the S-learner and PS-GLM tied at DGPs 32, 59, and 64.

The pattern of small differences also extends to the coverage and interval length, which were similarly close across all DGPs. PS-BART had on average slightly higher coverage than the other two variants, consistent with its comparatively wider confidence intervals. All three BART S-learners reached a coverage rate of 0.90 or above for all but three DGPs. As discussed in Section 6.3.1, DGP 30 and DGP 39 stand out with notably poor coverage. Coverage was at 0.95 or above in 8 cases for the S-learner, 9 cases for PS-GLM and 8 cases for PS-BART.

### **T-learner**

In most DGPs, the RMSE of the T-learner was close to that of the three S-learners. Most of the time it was slightly higher than all S-learners (e.g. in DGP 29, 58 and 61) or ranked somewhere between them (e.g. in DGP 32 or 39). Larger gaps to the S-learners appeared in DGPs 30, 37, 38, 40 and 62, with DGP30 showing the biggest difference of 0.058 to the S-learner and PS-GLM. The T-learner outperformed all three S-learners in only four cases (DGPs 57, 59, 60, 63), in three of them (all except 60) it had the lowest RMSE of all models. The differences, however, were very small, with 0.005 at DGP 57 being the largest difference to the S-learner, which placed second. The T-learner had a lower RMSE than the casual forest in 10 DGPs and a lower RMSE than AIPTW in 9 DGPs. In 11 DGPs, the T-learner reached a coverage rate of 0.90 or above, in 6 DGPs above 0.905. Comparing the coverage rates to the S-learners, reveals a similar pattern as the RMSE. For DGPs, where differences between T-learner and S-learners were small, coverage rates were also similar. Coverage was substantially lower than that of the S-learners at DGPs 30 (0.60), 37 (0.88), 38 (0.85) and 40 (0.73). Those are DGPs where noticeable differences in RMSE were present. Coverage at DGP 39 was also relatively low (0.83), however, here all S-learners showed similarly low or even lower coverage. The interval lengths of the T-learner did not deviate much from those of the S-learners.

### **Causal forest**

The causal forest had the highest RMSE in 9 cases, ranked 5th in 2 cases and 4th in 2 cases. In some cases (such as DGPs 29, 32, 60), there were no large differences to the model with the lowest RMSE (e.g. only 0.008 in DGP 32). However, in some DGPs, the RMSE of the causal forest was much higher than that of the best performing model. The notably

largest gaps were in DGP 37, a difference of 0.110 to AIPW and in DGP 39, a difference of 0.092 to AIPW and the S-learner. When the causal forest did not rank last, it was usually still worse than the S-learners and only outperformed the T-Learner (in DGPs 38) or both the T-learner and AIPW (in DGP 30, 62). There were two DGPs where the causal forest achieved the lowest RMSE out of all models (58 and 64) and one (63) where it ranked second and was outperformed by the T-learner by only 0.003. In all three cases, the difference to the next best model (in each case the S-learner) was very small (0.003 for DGP 58, 0.002 for DGP 63, and 0.002 for DGP 64).

The coverage of the causal forest was at 0.90 or above for 13 DGPs, in five of those at 0.95 or above 0.95, and slightly below 0.90 for DGP 38 (0.89) and DGP 60 (0.88), making its coverage the most stable among all models. The only outlier was DGP 30 (0.65), where the causal forest had a relatively high RMSE and most models had low coverage. The causal forest often had wider intervals than other models.

### **Augmented Inverse Probability Weighting**

AIPW had the highest RMSE in 4 DGPs and the second highest RMSE in 4 DGPs, often only underperformed by the causal forest. Differences to the best performing models were often moderate (e.g. 0.022 in DGP 62). Apart from DGP 30, the biggest difference to the best performing model was in DGP 64 (0.041 to the causal forest). It had the lowest RMSE in four DGPs (29, 37, 39, 60), however, in these DGPs all or most other models are very close.

The RMSE of AIPW at DGP 30 (1.196) stands out as an extreme outlier. The estimate was so far off that it resulted in a coverage of 0.00, which is by far the worst performance across the whole evaluation. Even though almost all other models, especially the T-learner and the causal forest, also had lower coverage rates for this DGP than on average, none of them came close to the failure of AIPW here. For the other DGPs, coverage was at 0.90 or above for 11 DGPs. Coverage rates were especially low at 40 (0.77) and 64 (0.81). CI Length did not deviate much from those of the BART models.

DGP 30 as an extreme outlier partly biased aggregated results. When excluding it from the calculation of the average metrics, would reduce the RMSE of AIPW from 0.171 to 0.102, while that of the causal forest would rise slightly from 0.117 to 0.119, making it the worst performing model instead. At the same time, AIPW's coverage would increase from 0.858 to 0.915, causing it to have a higher coverage than the T-learner (0.913).

## 6.4 Hyperparameter Tuning

The four DGPs that were chosen for hyperparameter tuning were DGPs 30, 40, 60 and 63. In DGP 30 and 40, the three S-learners achieved lower RMSE values than the other methods. In DGPs 60 and 63, the S-Learners were outperformed by other methods (AIPW and the T-learner in DGP 60 and the T-learner and causal forest in DGP 63).

It is important to note that the performance metrics of the default parameters can differ from those of the results in stage 1. This results from the fact that hyperparameter tuning was only performed on 20 datasets for each DGP. To ensure a fair comparison, the performance of the default specifications was recalculated using the same 20 datasets.

### 6.4.1 General Results

Overall, differences in performance between hyperparameter specifications were rather small. In many cases, the difference between the worst and the best performing specification were even smaller than the differences between methods.

For all BART models, the differences in RMSE between the best and the worst hyperparameter combinations was below 0.1 for DGPs 40 and 60. For DGP 30, only the T-learner exceeded 0.1 with a difference of 0.133 between the best (...) and worst combination (...), which is the largest difference among all BART models across all the included DGPs. The differences for DGP 63 were above 0.1 for all BART models, with X being highest at X. For the causal forest, the differences in the RMSE between the best and the worst hyperparameter combination were also rather small. Only for DGP 30, it exceeded 0.1 at 0.189. Since these relatively small performance differences, discussing all hyperparameter combinations in detail would provide limited insight. Therefore, only general patterns will be discussed. The detailed results for all hyperparameter specifications can be found in Appendix X.

When it comes to the best hyperparameters combinations, results were relatively consistent across the BART learners. For DGPs 30, 60 and 63, lower values of  $k$  clearly performed better than higher values, with RMSE constantly increasing from 1 to 5. For DGPs 60 and 63, these lower  $k$  values tended to work best in combination with a smaller numbers of trees, where 75 trees often outperformed 100 trees and 100 trees outperformed 200 trees. On the other hand, DGP 30 appeared to benefit from a larger number of trees. DGP 40 differed from the other settings, as higher values of  $k$  produced better results and a larger number of trees was better for the S-learner and PS-BART/BART. For the variance-related hyperparameters  $\nu$  and  $q$  no clear patterns emerged across DGPs and learners.

For the causal forest, a sample fraction of 0.5 generally yielded better results than a sample fraction of 0.3, with the exception of DGP 63. No consistent pattern emerged for *min.node.size*. For *mtry*, lower values lead to the best results for DGPs 30 and 40, whereas higher values were preferred for DGP 60 and 63 (the highest being  $\sqrt{p} + 20$ ).

IPTW TODO!

#### 6.4.2 Improvements by DGP

Apart from evaluating which hyperparameters yield the best results overall, it is also evaluated to which extent tuning improved model performance. To do so, the best performing hyperparameter specification is compared with the default setting. Since many of the improvements were rather small, only the most notable changes are discussed. Figures X-X show the RMSE differences between the default and tuned specifications. Tables X-X (TODO) provide a more detailed overview.

For DGP 30, hyperparameter tuning had little impact on the BART models and the causal forest. The most substantial improvement happened for AIPW, where estimating the propensity score with BART reduced the RMSE from 1.186 to 0.140. This is the largest improvement across all methods and DGPs. However, AIPW still remains the worst performing method. The ranking of the methods only changes for the T-learner and the causal forest, with the T-learner slightly outperforming the causal forest when tuned. Coverage increased by 0.05 for PS-BART and by 0.10 for the S-learner and T-learner each. The largest increase was again observed for AIPW with an increase from 0.00 to 0.70.

For DGP 40, hyperparameter tuning had virtually no impact on model performance. The largest decrease in RMSE could be observed at 0.005 for AIPW and the T-learner. For the causal forest, the default parameters actually were the best specification. Even though the S-learner only improved its performance by 0.001, it became the sole best performing model. In the default setting, it had shared the first place with PS-GLM. Coverage of the T-learner and AIPW increased by 0.05 each.

For DGP 60, tuning again only resulted in small performance improvements. The largest RMSE reduction was 0.004 for PS-GLM, followed by S-learner and causal forest with 0.003. These improvements resulted in both PS-GLM and the S-learner passing PS-BART. Coverage increased by 0.05 for PS-BART/GLM and PS-BART/BART.

For DGP 63, the causal forest had the largest RMSE reduction of 0.023, followed by PS-GLM and the S-learner with 0.013 each. The S-learner passed the T-learner and became the best performing method. Coverage increased by 0.05 for S-learner, PS-GLM and PS-BART and by 0.10 for AIPW.





## **7 Fazit und Ausblick**

(...)



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## Results

Tabelle .1: Relative RMSE, coverage and relative interval length by model and DGP

| Model          | DGP    | RMSE  | Coverage | Interval Length |
|----------------|--------|-------|----------|-----------------|
| BART/S-learner | DGP 29 | 0.109 | 0.90     | 0.380           |
| BART/T-learner | DGP 29 | 0.116 | 0.91     | 0.392           |
| BART/PS-GLM    | DGP 29 | 0.109 | 0.92     | 0.381           |
| BART/PS-BART   | DGP 29 | 0.108 | 0.94     | 0.382           |
| Causal forest  | DGP 29 | 0.116 | 0.92     | 0.414           |
| AIPW           | DGP 29 | 0.107 | 0.92     | 0.386           |
| BART/S-learner | DGP 30 | 0.058 | 0.84     | 0.181           |
| BART/T-learner | DGP 30 | 0.102 | 0.60     | 0.203           |
| BART/PS-GLM    | DGP 30 | 0.058 | 0.86     | 0.181           |
| BART/PS-BART   | DGP 30 | 0.049 | 0.94     | 0.186           |
| Causal forest  | DGP 30 | 0.093 | 0.65     | 0.195           |
| AIPW           | DGP 30 | 1.196 | 0.00     | 0.228           |
| BART/S-learner | DGP 31 | 0.193 | 0.95     | 0.764           |
| BART/T-learner | DGP 31 | 0.194 | 0.98     | 0.790           |
| BART/PS-GLM    | DGP 31 | 0.196 | 0.95     | 0.799           |
| BART/PS-BART   | DGP 31 | 0.202 | 0.95     | 0.806           |
| Causal forest  | DGP 31 | 0.248 | 0.92     | 0.904           |
| AIPW           | DGP 31 | 0.219 | 0.95     | 0.815           |
| BART/S-learner | DGP 32 | 0.124 | 0.99     | 0.528           |
| BART/T-learner | DGP 32 | 0.124 | 0.99     | 0.532           |
| BART/PS-GLM    | DGP 32 | 0.124 | 0.99     | 0.529           |
| BART/PS-BART   | DGP 32 | 0.128 | 0.97     | 0.543           |
| Causal forest  | DGP 32 | 0.132 | 0.97     | 0.527           |
| AIPW           | DGP 32 | 0.127 | 0.98     | 0.526           |

Tabelle .2: Relative RMSE, coverage and relative interval length by model and DGP

| Model          | DGP    | RMSE  | Coverage | Interval Length |
|----------------|--------|-------|----------|-----------------|
| BART/S-learner | DGP 37 | 0.052 | 0.96     | 0.199           |
| BART/T-learner | DGP 37 | 0.070 | 0.88     | 0.230           |
| BART/PS-GLM    | DGP 37 | 0.052 | 0.95     | 0.199           |
| BART/PS-BART   | DGP 37 | 0.051 | 0.97     | 0.202           |
| Causal forest  | DGP 37 | 0.149 | 0.92     | 0.494           |
| AIPW           | DGP 37 | 0.039 | 0.91     | 0.136           |
| BART/S-learner | DGP 38 | 0.056 | 0.97     | 0.225           |
| BART/T-learner | DGP 38 | 0.074 | 0.85     | 0.222           |
| BART/PS-GLM    | DGP 38 | 0.057 | 0.98     | 0.230           |
| BART/PS-BART   | DGP 38 | 0.059 | 0.99     | 0.236           |
| Causal forest  | DGP 38 | 0.062 | 0.89     | 0.211           |
| AIPW           | DGP 38 | 0.059 | 0.86     | 0.182           |
| BART/S-learner | DGP 39 | 0.053 | 0.83     | 0.145           |
| BART/T-learner | DGP 39 | 0.057 | 0.83     | 0.158           |
| BART/PS-GLM    | DGP 39 | 0.056 | 0.81     | 0.151           |
| BART/PS-BART   | DGP 39 | 0.058 | 0.77     | 0.153           |
| Causal forest  | DGP 39 | 0.145 | 0.93     | 0.505           |
| AIPW           | DGP 39 | 0.053 | 0.95     | 0.242           |
| BART/S-learner | DGP 40 | 0.020 | 0.94     | 0.077           |
| BART/T-learner | DGP 40 | 0.038 | 0.73     | 0.086           |
| BART/PS-GLM    | DGP 40 | 0.020 | 0.95     | 0.077           |
| BART/PS-BART   | DGP 40 | 0.020 | 0.96     | 0.078           |
| Causal forest  | DGP 40 | 0.069 | 0.98     | 0.259           |
| AIPW           | DGP 40 | 0.051 | 0.77     | 0.130           |

Tabelle .3: Relative RMSE, coverage and relative interval length by model and DGP

| Model          | DGP    | RMSE  | Coverage | Interval Length |
|----------------|--------|-------|----------|-----------------|
| BART/S-learner | DGP 57 | 0.168 | 0.95     | 0.706           |
| BART/T-learner | DGP 57 | 0.163 | 0.96     | 0.704           |
| BART/PS-GLM    | DGP 57 | 0.170 | 0.95     | 0.710           |
| BART/PS-BART   | DGP 57 | 0.173 | 0.94     | 0.713           |
| Causal forest  | DGP 57 | 0.194 | 0.95     | 0.712           |
| AIPW           | DGP 57 | 0.174 | 0.96     | 0.713           |
| BART/S-learner | DGP 58 | 0.107 | 0.93     | 0.392           |
| BART/T-learner | DGP 58 | 0.111 | 0.92     | 0.404           |
| BART/PS-GLM    | DGP 58 | 0.109 | 0.92     | 0.394           |
| BART/PS-BART   | DGP 58 | 0.110 | 0.94     | 0.397           |
| Causal forest  | DGP 58 | 0.104 | 0.92     | 0.366           |
| AIPW           | DGP 58 | 0.110 | 0.92     | 0.388           |
| BART/S-learner | DGP 59 | 0.109 | 0.94     | 0.458           |
| BART/T-learner | DGP 59 | 0.106 | 0.95     | 0.453           |
| BART/PS-GLM    | DGP 59 | 0.109 | 0.98     | 0.469           |
| BART/PS-BART   | DGP 59 | 0.110 | 0.96     | 0.474           |
| Causal forest  | DGP 59 | 0.123 | 0.92     | 0.438           |
| AIPW           | DGP 59 | 0.115 | 0.97     | 0.468           |
| BART/S-learner | DGP 60 | 0.054 | 0.89     | 0.179           |
| BART/T-learner | DGP 60 | 0.049 | 0.92     | 0.178           |
| BART/PS-GLM    | DGP 60 | 0.053 | 0.90     | 0.180           |
| BART/PS-BART   | DGP 60 | 0.053 | 0.90     | 0.181           |
| Causal forest  | DGP 60 | 0.058 | 0.88     | 0.180           |
| AIPW           | DGP 60 | 0.048 | 0.93     | 0.181           |

Tabelle .4: Relative RMSE, coverage and relative interval length by model and DGP

| Model          | DGP    | RMSE  | Coverage | Interval Length |
|----------------|--------|-------|----------|-----------------|
| BART/S-learner | DGP 61 | 0.087 | 0.98     | 0.413           |
| BART/T-learner | DGP 61 | 0.100 | 0.95     | 0.418           |
| BART/PS-GLM    | DGP 61 | 0.088 | 0.98     | 0.417           |
| BART/PS-BART   | DGP 61 | 0.088 | 0.98     | 0.420           |
| Causal forest  | DGP 61 | 0.105 | 0.97     | 0.415           |
| AIPW           | DGP 61 | 0.094 | 0.98     | 0.424           |
| BART/S-learner | DGP 62 | 0.105 | 0.96     | 0.429           |
| BART/T-learner | DGP 62 | 0.124 | 0.95     | 0.434           |
| BART/PS-GLM    | DGP 62 | 0.105 | 0.96     | 0.435           |
| BART/PS-BART   | DGP 62 | 0.105 | 0.95     | 0.440           |
| Causal forest  | DGP 62 | 0.114 | 0.93     | 0.417           |
| AIPW           | DGP 62 | 0.127 | 0.93     | 0.433           |
| BART/S-learner | DGP 63 | 0.120 | 0.96     | 0.459           |
| BART/T-learner | DGP 63 | 0.115 | 0.94     | 0.480           |
| BART/PS-GLM    | DGP 63 | 0.125 | 0.94     | 0.485           |
| BART/PS-BART   | DGP 63 | 0.129 | 0.94     | 0.492           |
| Causal forest  | DGP 63 | 0.118 | 0.90     | 0.427           |
| AIPW           | DGP 63 | 0.130 | 0.88     | 0.399           |
| BART/S-learner | DGP 64 | 0.044 | 0.94     | 0.170           |
| BART/T-learner | DGP 64 | 0.046 | 0.93     | 0.167           |
| BART/PS-GLM    | DGP 64 | 0.044 | 0.93     | 0.171           |
| BART/PS-BART   | DGP 64 | 0.046 | 0.93     | 0.173           |
| Causal forest  | DGP 64 | 0.042 | 0.95     | 0.170           |
| AIPW           | DGP 64 | 0.083 | 0.81     | 0.207           |

Tabelle .5: Ranking of models by RMSE across all 16 DGPs (rank 1 = lowest RMSE)

| Model          | 1st | 2nd | 3rd | 4th | 5th | 6th | Mean Rank |
|----------------|-----|-----|-----|-----|-----|-----|-----------|
| BART/S-learner | 7   | 5   | 3   | 0   | 1   | 0   | 1.94      |
| BART/PS-GLM    | 3   | 5   | 7   | 1   | 0   | 0   | 2.38      |
| BART/PS-BART   | 3   | 3   | 2   | 5   | 3   | 0   | 3.12      |
| BART/T-learner | 4   | 2   | 0   | 3   | 5   | 2   | 3.56      |
| AIPW           | 4   | 0   | 1   | 3   | 4   | 4   | 3.94      |
| Causal forest  | 2   | 1   | 0   | 2   | 2   | 9   | 4.75      |

Tabelle .6: Ranking of models by coverage across all 16 DGPs (rank 1 = highest coverage)

| Model          | 1st | 2nd | 3rd | 4th | 5th | 6th | Mean Rank |
|----------------|-----|-----|-----|-----|-----|-----|-----------|
| BART/PS-GLM    | 4   | 5   | 6   | 0   | 1   | 0   | 2.31      |
| BART/PS-BART   | 6   | 3   | 4   | 0   | 1   | 2   | 2.56      |
| BART/S-learner | 4   | 4   | 4   | 1   | 2   | 1   | 2.75      |
| AIPW           | 4   | 3   | 1   | 1   | 4   | 3   | 3.44      |
| BART/T-learner | 3   | 2   | 4   | 1   | 2   | 4   | 3.56      |
| Causal forest  | 2   | 2   | 2   | 3   | 4   | 3   | 3.88      |

Tabelle .7: Ranking of models by interval length across all 16 DGPs (rank 1 = shortest interval)

| Model          | 1st | 2nd | 3rd | 4th | 5th | 6th | Mean Rank |
|----------------|-----|-----|-----|-----|-----|-----|-----------|
| BART/S-learner | 6   | 5   | 4   | 1   | 0   | 0   | 2.00      |
| BART/PS-GLM    | 2   | 3   | 4   | 3   | 4   | 0   | 3.25      |
| Causal forest  | 3   | 5   | 1   | 2   | 0   | 5   | 3.38      |
| BART/T-learner | 3   | 2   | 1   | 5   | 4   | 1   | 3.50      |
| AIPW           | 4   | 1   | 1   | 2   | 5   | 3   | 3.75      |
| BART/PS-BART   | 0   | 0   | 4   | 2   | 5   | 5   | 4.69      |



## Tuning tables

| Model          | Setting             | k | n_trees | $\nu$ | q    | min.node.size | sample.fraction | mtry | RMSE (rel.) | Coverage | CI length (rel.) |
|----------------|---------------------|---|---------|-------|------|---------------|-----------------|------|-------------|----------|------------------|
| BART/S-learner | default             | 2 | 75      | 3     | 0.90 | –             | –               | –    | 0.064       | 0.75     | 0.179            |
| BART/S-learner | tuned               | 1 | 200     | 10    | 0.75 | –             | –               | –    | 0.059       | 0.85     | 0.179            |
| BART/T-learner | default             | 2 | 75      | 3     | 0.90 | –             | –               | –    | 0.110       | 0.50     | 0.200            |
| BART/T-learner | tuned               | 1 | 200     | 10    | 0.75 | –             | –               | –    | 0.090       | 0.60     | 0.202            |
| PS-BART/GLM    | default             | 2 | 75      | 3     | 0.90 | –             | –               | –    | 0.065       | 0.75     | 0.180            |
| PS-BART/GLM    | tuned               | 1 | 200     | 3     | 0.99 | –             | –               | –    | 0.060       | 0.80     | 0.178            |
| PS-BART/BART   | default             | 2 | 75      | 3     | 0.90 | –             | –               | –    | 0.048       | 0.95     | 0.184            |
| PS-BART/BART   | tuned               | 3 | 100     | 10    | 0.75 | –             | –               | –    | 0.047       | 0.95     | 0.183            |
| Causal forest  | default             | – | –       | –     | –    | 5             | 0.5             | 26   | 0.100       | 0.55     | 0.193            |
| Causal forest  | tuned               | – | –       | –     | –    | 20            | 0.5             | 26   | 0.096       | 0.55     | 0.194            |
| AIPW           | Logistic regression | – | –       | –     | –    | –             | –               | –    | 1.191       | 0.00     | 0.225            |
| AIPW           | BART                | – | –       | –     | –    | –             | –               | –    | 0.139       | 0.70     | 0.298            |

Tabelle .8: DGP 30: Performance comparison of default and tuned model specifications.

| Model          | Setting             | k | n_trees | $\nu$ | q    | min.node.size | sample.fraction | mtry | RMSE (rel.) | Coverage | CI length (rel.) |
|----------------|---------------------|---|---------|-------|------|---------------|-----------------|------|-------------|----------|------------------|
| BART/S-learner | tuned               | 5 | 200     | 10    | 0.75 | –             | –               | –    | 0.019       | 0.95     | 0.080            |
| BART/S-learner | tuned               | 5 | 100     | 3     | 0.99 | –             | –               | –    | 0.019       | 0.95     | 0.076            |
| BART/T-learner | tuned               | 5 | 100     | 3     | 0.99 | –             | –               | –    | 0.018       | 0.95     | 0.075            |
| BART/T-learner | tuned               | 5 | 75      | 3     | 0.99 | –             | –               | –    | 0.031       | 0.90     | 0.112            |
| PS-BART/GLM    | default             | 2 | 75      | 3     | 0.90 | –             | –               | –    | 0.019       | 0.95     | 0.078            |
| PS-BART/GLM    | default             | 2 | 75      | 3     | 0.90 | –             | –               | –    | 0.036       | 0.85     | 0.088            |
| PS-BART/BART   | default             | 2 | 75      | 3     | 0.90 | –             | –               | –    | 0.020       | 0.95     | 0.079            |
| PS-BART/BART   | default             | 2 | 75      | 3     | 0.90 | –             | –               | –    | 0.019       | 0.95     | 0.077            |
| Causal forest  | default             | – | –       | –     | –    | 5             | 0.5             | 22   | 0.055       | 1.00     | 0.228            |
| Causal forest  | tuned               | – | –       | –     | –    | 5             | 0.5             | 22   | 0.055       | 1.00     | 0.228            |
| AIPW           | Logistic regression | – | –       | –     | –    | –             | –               | –    | 0.043       | 0.80     | 0.126            |
| AIPW           | BART                | – | –       | –     | –    | –             | –               | –    | 0.037       | 0.90     | 0.119            |

Tabelle .9: DGP 40: Performance comparison of default and tuned model specifications.

| Model          | Setting             | k | n_trees | $\nu$ | q    | min.node.size | sample.fraction | mtry | RMSE (rel.) | Coverage | CI length (rel.) |
|----------------|---------------------|---|---------|-------|------|---------------|-----------------|------|-------------|----------|------------------|
| BART/S-learner | default             | 2 | 75      | 3     | 0.90 | –             | –               | –    | 0.062       | 0.90     | 0.179            |
| BART/S-learner | tuned               | 1 | 75      | 3     | 0.90 | –             | –               | –    | 0.059       | 0.95     | 0.182            |
| BART/T-learner | default             | 2 | 75      | 3     | 0.90 | –             | –               | –    | 0.053       | 0.95     | 0.177            |
| BART/T-learner | tuned               | 1 | 75      | 3     | 0.99 | –             | –               | –    | 0.052       | 0.95     | 0.181            |
| PS-BART/GLM    | default             | 2 | 75      | 3     | 0.90 | –             | –               | –    | 0.062       | 0.90     | 0.180            |
| PS-BART/GLM    | tuned               | 1 | 75      | 3     | 0.90 | –             | –               | –    | 0.058       | 0.95     | 0.182            |
| PS-BART/BART   | default             | 2 | 75      | 3     | 0.90 | –             | –               | –    | 0.061       | 0.95     | 0.185            |
| PS-BART/BART   | tuned               | 2 | 75      | 10    | 0.75 | –             | –               | –    | 0.060       | 0.95     | 0.184            |
| Causal forest  | default             | – | –       | –     | –    | 5             | 0.5             | 26   | 0.063       | 0.85     | 0.179            |
| Causal forest  | tuned               | – | –       | –     | –    | 5             | 0.5             | 11   | 0.060       | 0.85     | 0.180            |
| AIPW           | Logistic Regression | – | –       | –     | –    | –             | –               | –    | 0.054       | 0.95     | 0.181            |
| AIPW           | BART                | – | –       | –     | –    | –             | –               | –    | 0.054       | 0.95     | 0.177            |

Tabelle .10: DGP 60: Performance comparison of default and tuned model specifications.

| Model          | Setting             | k | n.trees | $\nu$ | q    | min.node.size | sample.fraction | mtry | RMSE (rel.) | Coverage | CI length (rel.) |
|----------------|---------------------|---|---------|-------|------|---------------|-----------------|------|-------------|----------|------------------|
| BART/S-learner | default             | 2 | 75      | 3     | 0.90 | –             | –               | –    | 0.112       | 0.95     | 0.458            |
| BART/S-learner | tuned               | 1 | 75      | 10    | 0.75 | –             | –               | –    | 0.099       | 1.00     | 0.467            |
| BART/T-learner | default             | 2 | 75      | 3     | 0.90 | –             | –               | –    | 0.105       | 0.95     | 0.479            |
| BART/T-learner | tuned               | 3 | 75      | 3     | 0.99 | –             | –               | –    | 0.103       | 0.95     | 0.459            |
| PS-BART/GLM    | default             | 2 | 75      | 3     | 0.90 | –             | –               | –    | 0.121       | 0.95     | 0.487            |
| PS-BART/GLM    | tuned               | 1 | 75      | 3     | 0.90 | –             | –               | –    | 0.108       | 1.00     | 0.498            |
| PS-BART/BART   | default             | 2 | 75      | 3     | 0.90 | –             | –               | –    | 0.127       | 0.95     | 0.497            |
| PS-BART/BART   | tuned               | 1 | 75      | 3     | 0.99 | –             | –               | –    | 0.117       | 1.00     | 0.525            |
| Causal forest  | default             | – | –       | –     | –    | 5             | 0.5             | 26   | 0.144       | 0.80     | 0.431            |
| Causal forest  | tuned               | – | –       | –     | –    | 50            | 0.3             | 10   | 0.121       | 0.80     | 0.367            |
| AIPW           | Logistic regression | – | –       | –     | –    | –             | –               | –    | 0.111       | 0.95     | 0.401            |
| AIPW           | BART                | – | –       | –     | –    | –             | –               | –    | 0.155       | 0.95     | 0.478            |

Tabelle .11: DGP 63: Performance comparison of default and tuned model specifications.

Hiermit erkläre ich an Eides statt, dass ich die vorliegende Arbeit selbständig und ohne unerlaubte fremde Hilfe angefertigt, andere als die angegebenen Quellen und Hilfsmittel nicht benutzt und die den benutzten Quellen und Hilfsmitteln wörtlich oder inhaltlich entnommenen Stellen als solche kenntlich gemacht habe.

Bamberg, den xx. Mai 2099



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*(Unterschrift des Verfassers)*